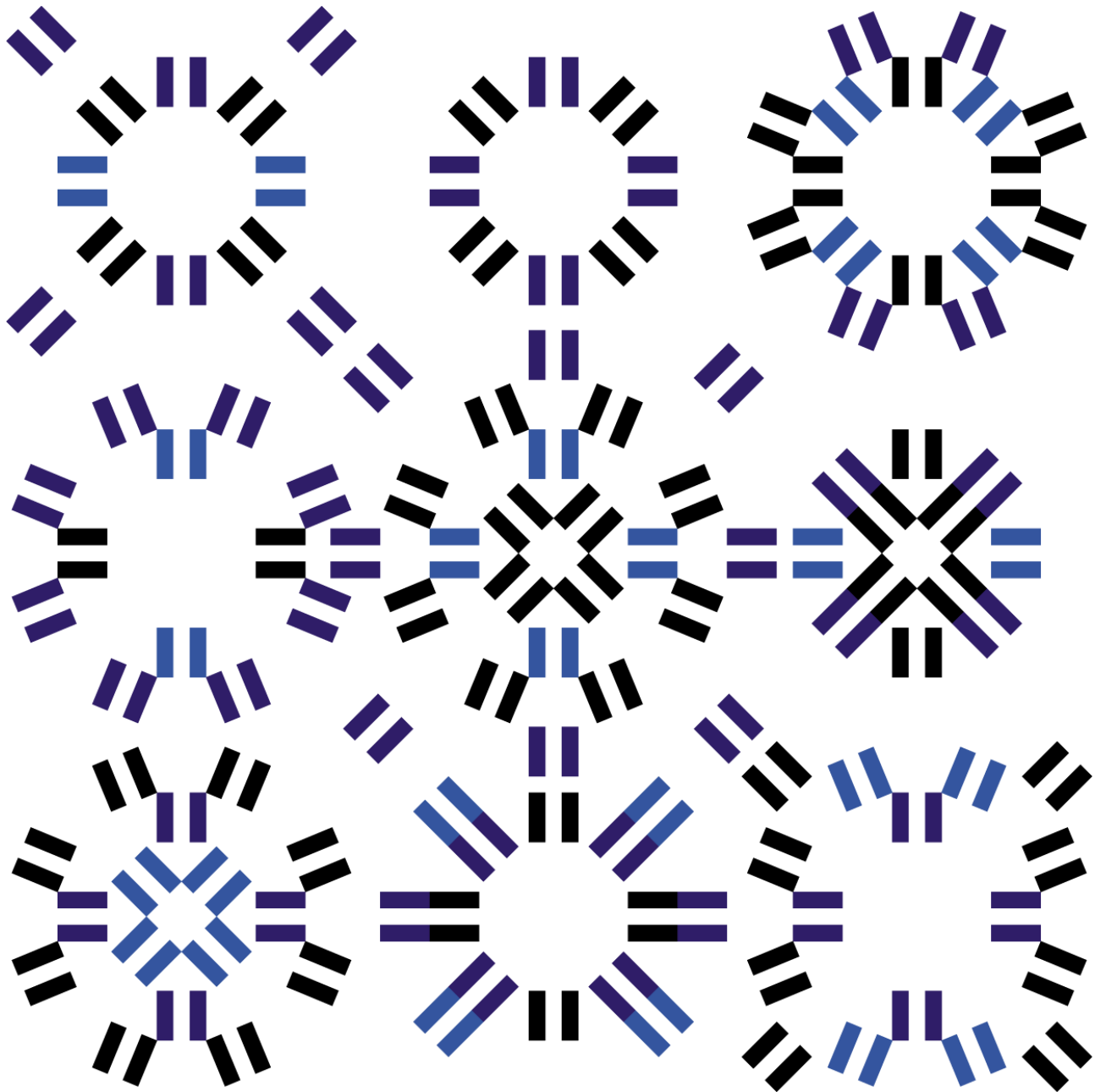
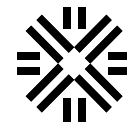




Big Data / Data Bases

Data Caching

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2026-2027



Université
de Rennes

Introducing example

Process of preparing and serving coffee at a café (1)

- Without caching
 - Customer 1 orders a coffee
 - Barista grinds the beans
 - Barista brews the coffee
 - Barista serves the coffee
 - Customer 2 orders a similar coffee
 - Barista grinds the beans
 - Barista brews the coffee
 - Barista serves the coffee

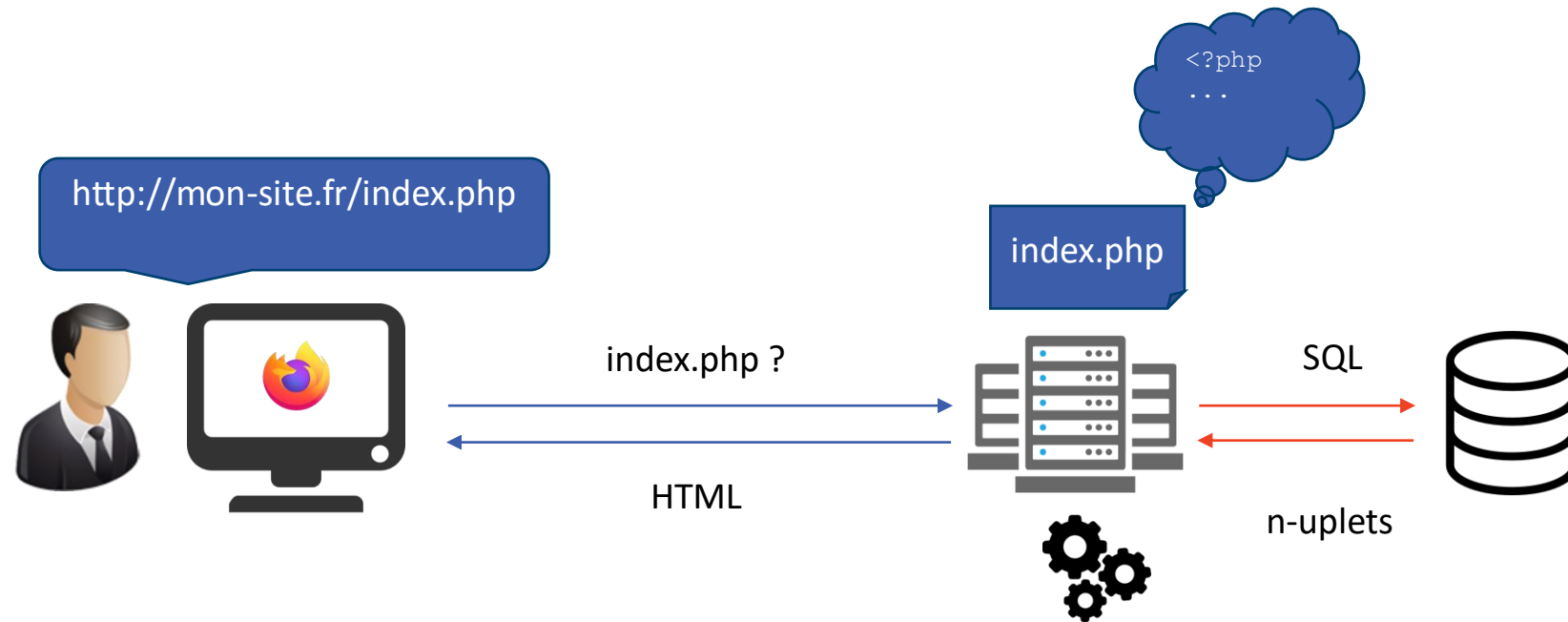
Introducing example

Process of preparing and serving coffee at a café (2)

- With caching
 - The café decides to prepare a pot of coffee in advance
 - Barista grinds the beans
 - Barista brews the coffee
 - (and keep it in a thermal carafe).
 - Customer 1 orders a coffee
 - Barista serves the coffee
 - Customer 2 orders a similar coffee
 - Barista serves the coffee

Introducing example

Web app. – Basic flow in a 3-tier architecture

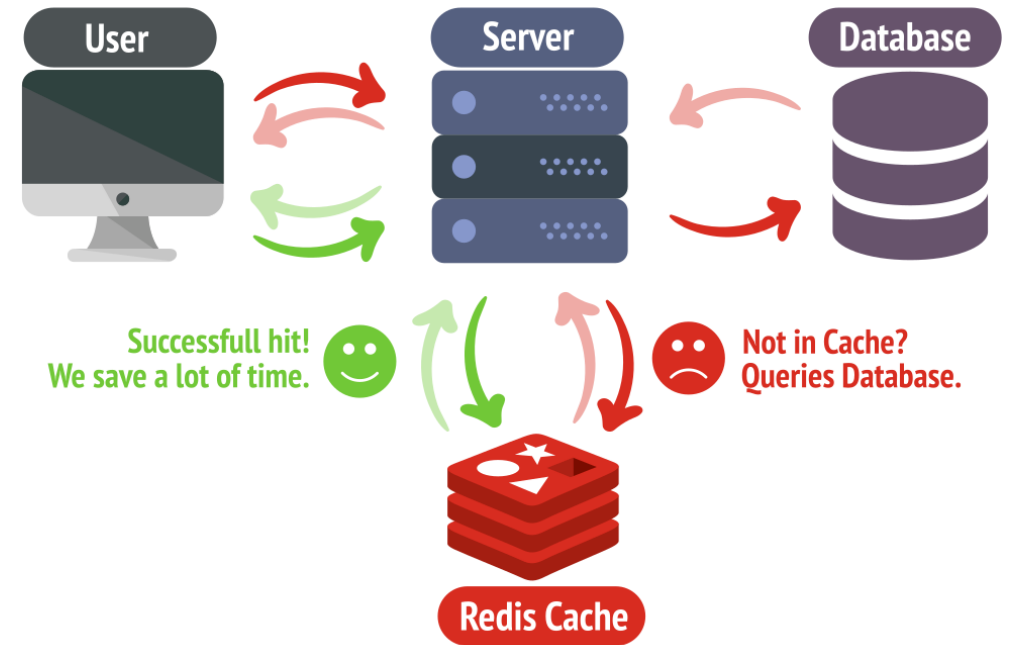


Code exécuté sur le serveur

Introducing example

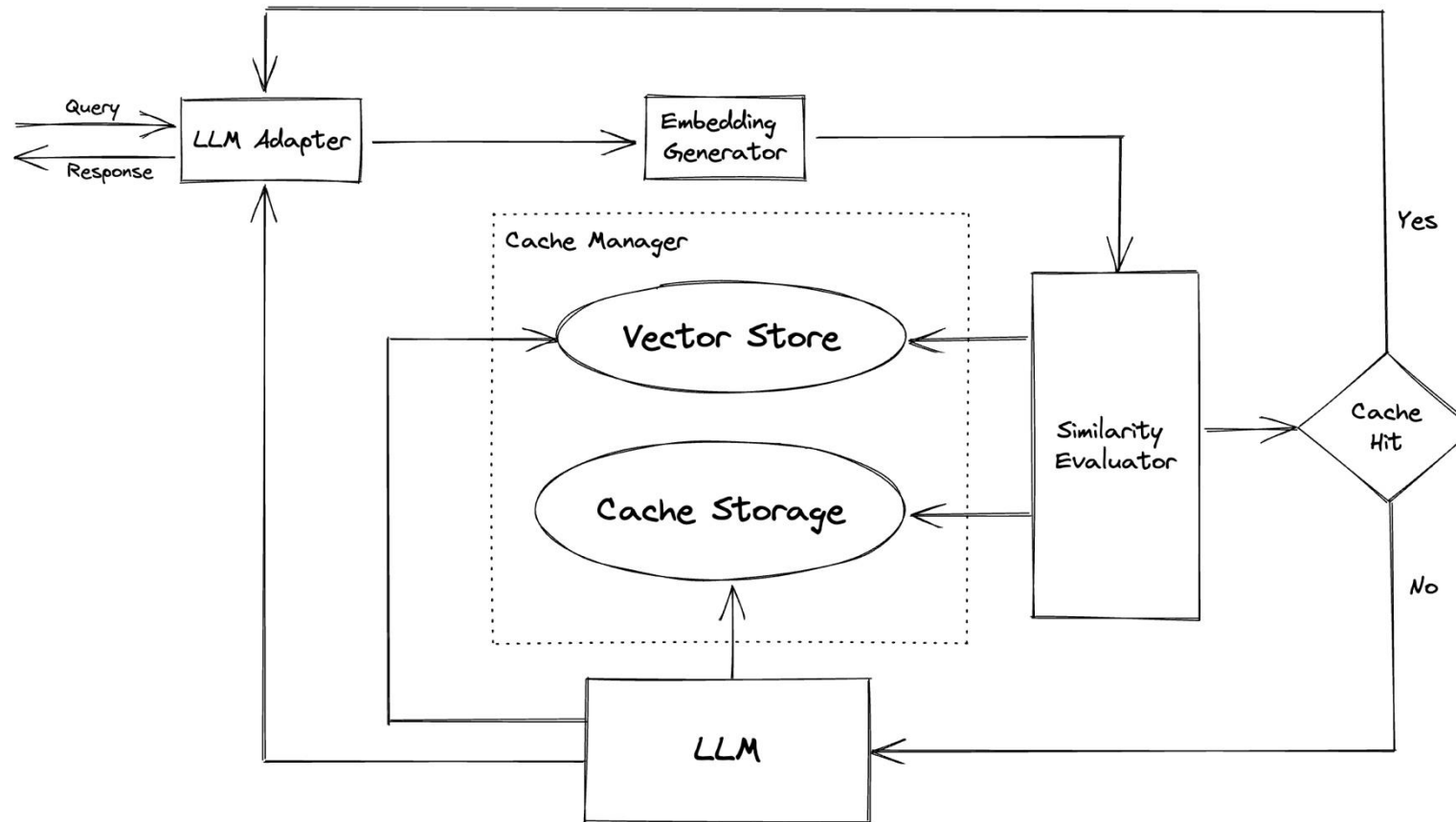
Web app. – Redis caching

- Client request to the PHP server to fetch data
- Data cached in Redis are checked by the PHP application
- Case of cache miss (data not found)
 - Data fetched/retrieved from the (SQL) DB
 - Data stored in the cache
- Case of cache hit
 - Data fetched from the cache
- Data returned to the user



Source: medium.com

Introducing example LLM caching



Source: <https://github.com/zilliztech/GPTCache>

Outline

- I. Motivation and principles
- II. Concepts
- III. Semantic caching
- IV. Cooperative caching

Outline

- I. Motivation and principles
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- III. Semantic caching
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Motivation and principles

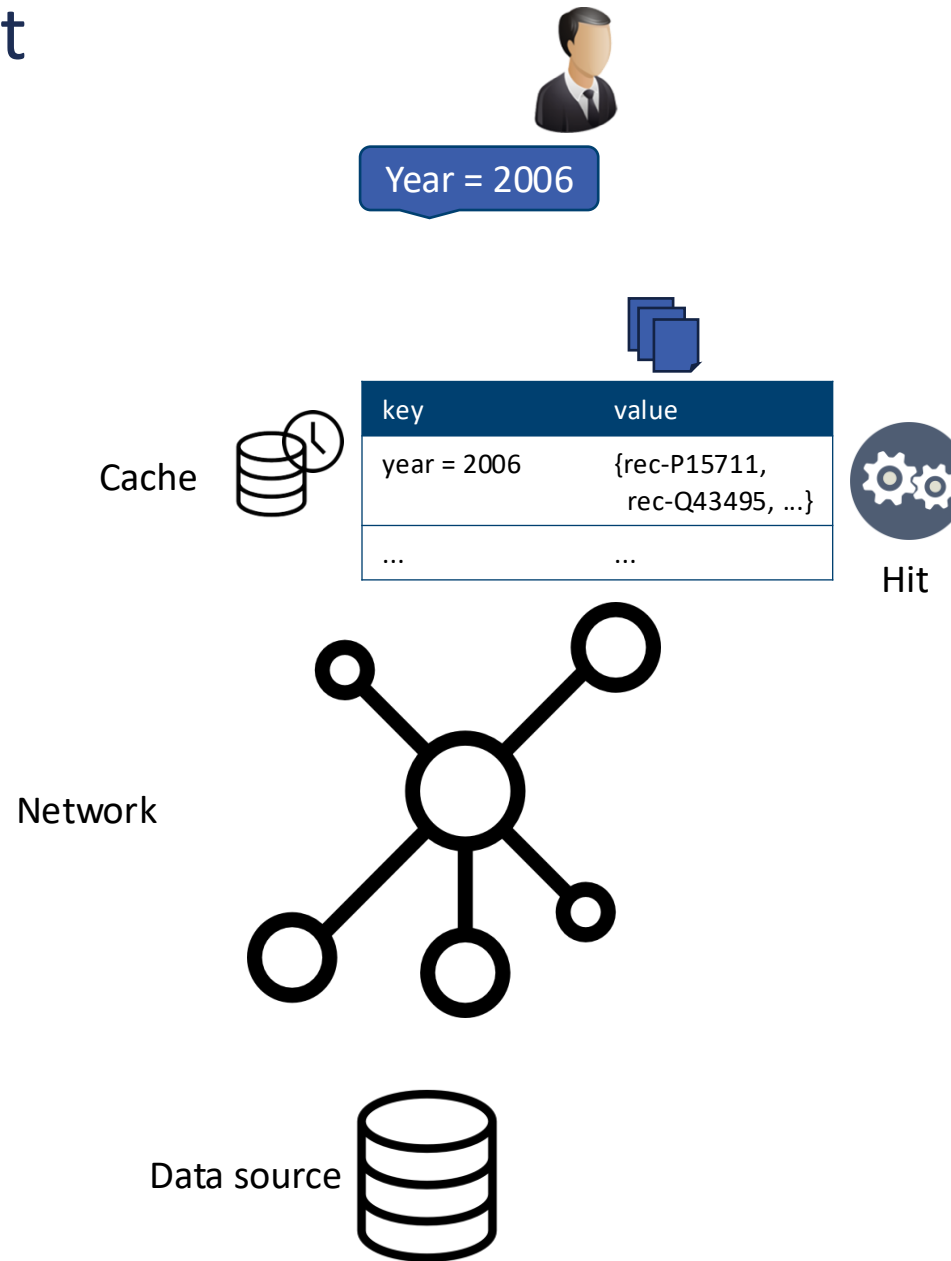
Definition and applications

- Definition
 - *Physical space with rapid access, generally of limited size, used to store copies of the items most likely to be referenced.* [LDO07]
- Applications
 - Focus
 - Web [BarishIEEEComMag00, BraunCompNet95] and DB caching
 - Other use cases
 - CPU cache
 - Disk cache
 - DNS cache
 - Etc.

Motivation and principles

Cache benefits – cache hit

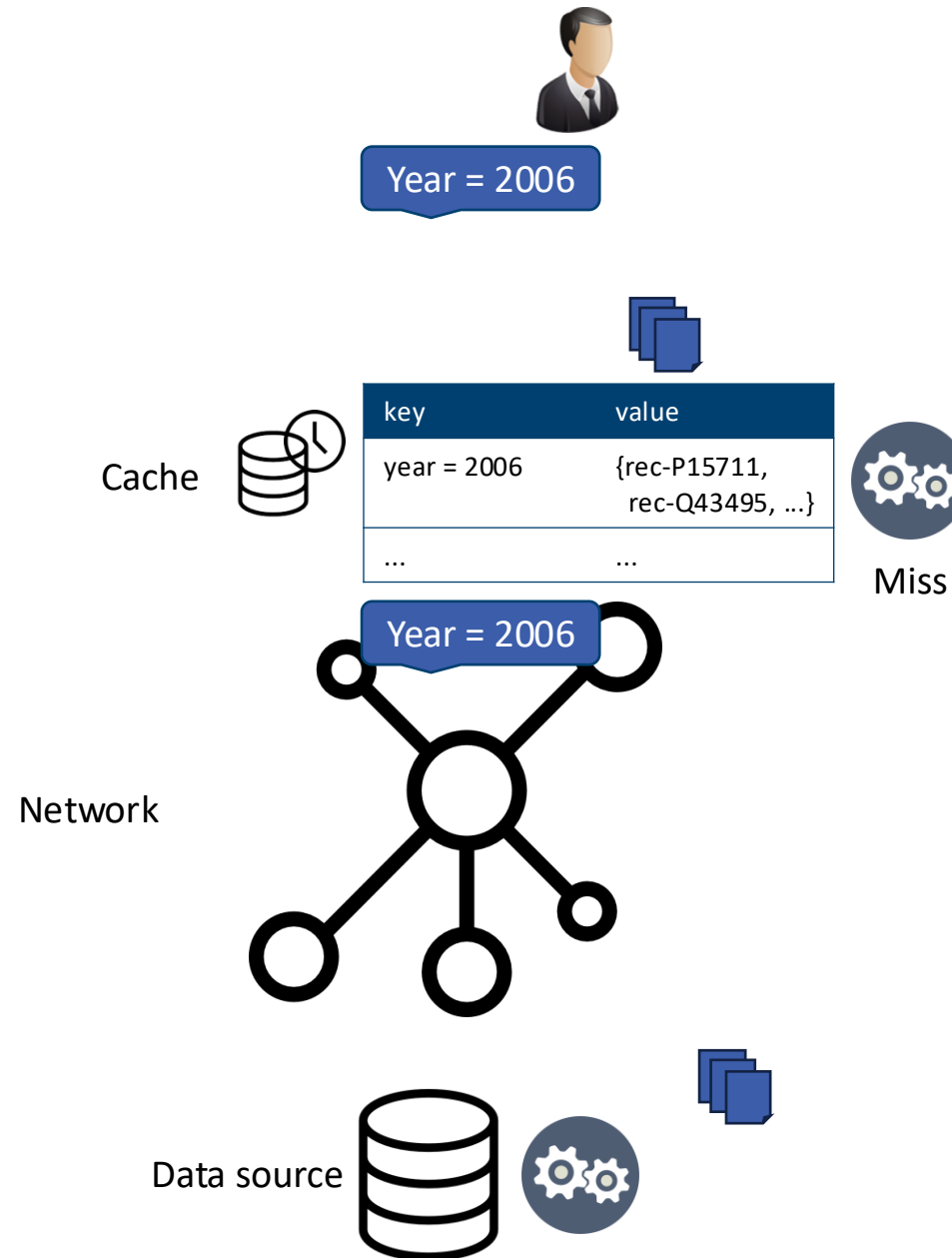
- (Monetary cost reduction)
- Performance optimization
 - Response time
 - Physical support
 - Location
 - Load reduction
 - On servers (app and DB)
 - On networks
- Availability



Motivation and principles

Cache miss

- (Monetary cost reduction)
- Performance optimization
 - Response time
 - Physical support
 - Location
 - Load reduction
 - On servers (app and DB)
 - On networks
- Availability



Outline

- I. Motivation and principles
- II. Concepts
 - A. Deployment
 - B. Basic functionalities
 - C. Optional functionalities
- III. Semantic caching
- IV. Cooperative caching

A – Deployment

Underlying systems (case of LLMs)

- Definition
 - System to be used to store the response
- Alternatives
 - Relational DBMS: SQLite, PostgreSQL, MySQL, MariaDB, etc.
 - Document store: DynamoDB

A – Deployment

Physical storage support

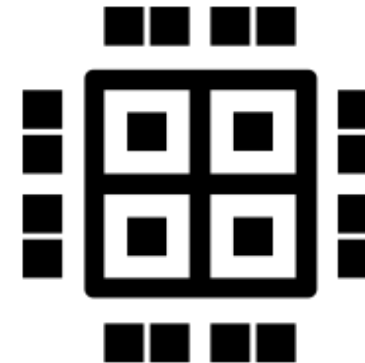
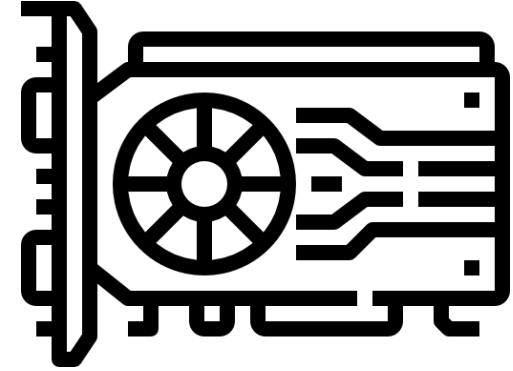
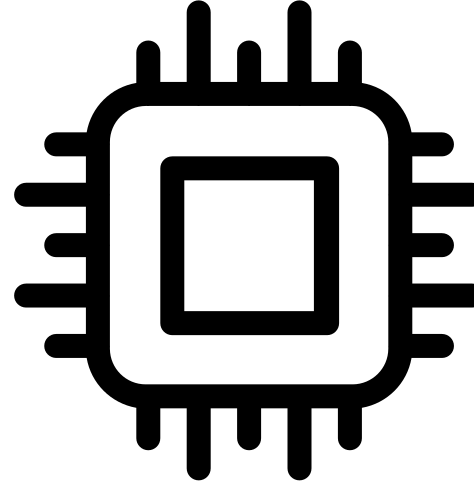
- Definition
 - Hardware used to deploy the cache
- Values
 - Memory
 - Fast access
 - Disk
 - Large storage
 - Persistency
 - Hybrid (cooperative caching)



A – Deployment

Physical processing

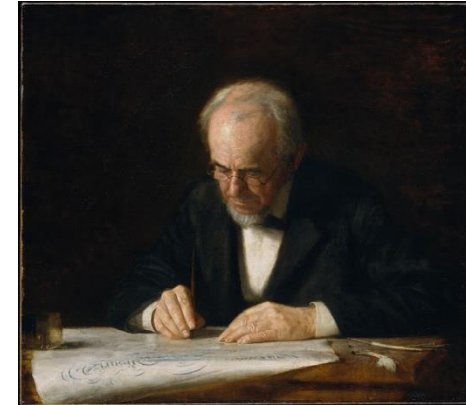
- Definition
 - Processing unit used to manage the cache
- Values
 - CPU
 - GPU
 - FPGA [dL20]



A – Deployment

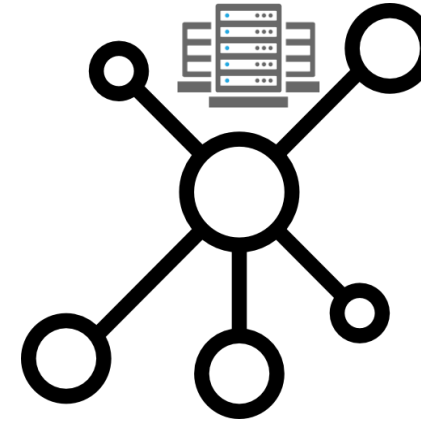
Access right

- Definition
 - Right to access an element of the cache
- Values
 - Read right
 - Write right
- N.B.
 - Access right can be
 - Global to a cache
 - User-specific
 - Document-specific



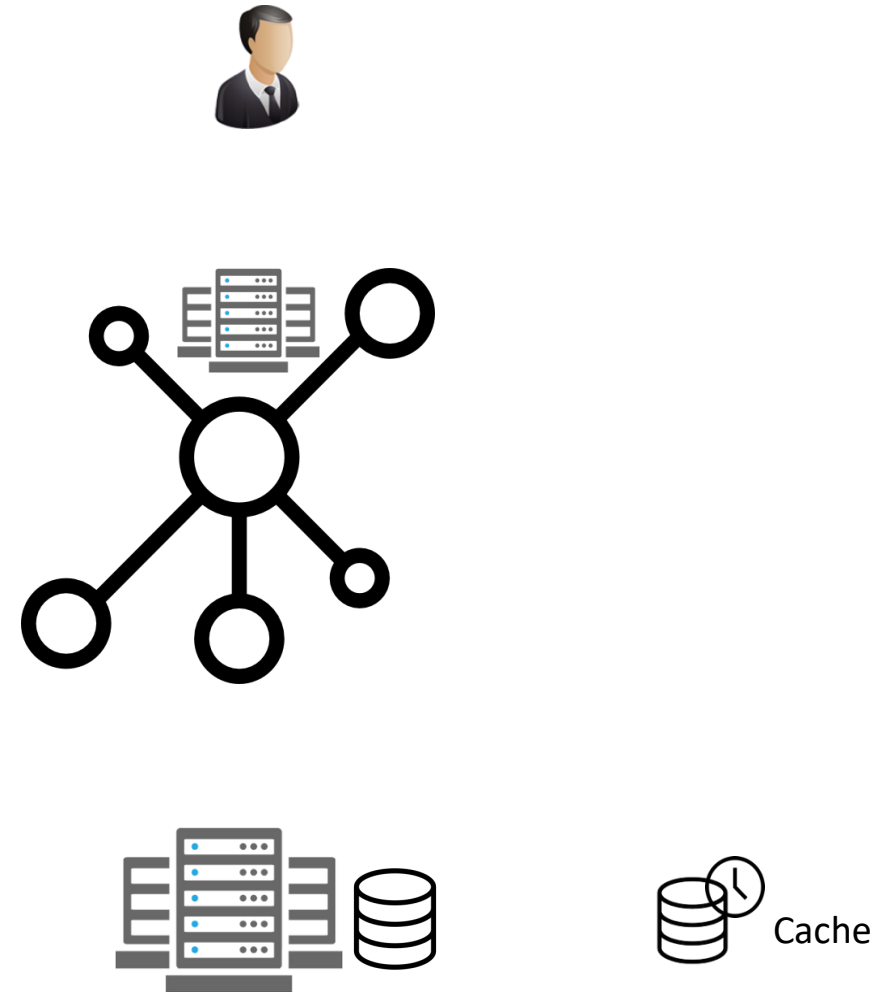
A – Deployment Location

- Definition
 - Host of the cache
- Values
 - + Availability
 - + Load balancing
 - + Scalability
 - + User-driven content



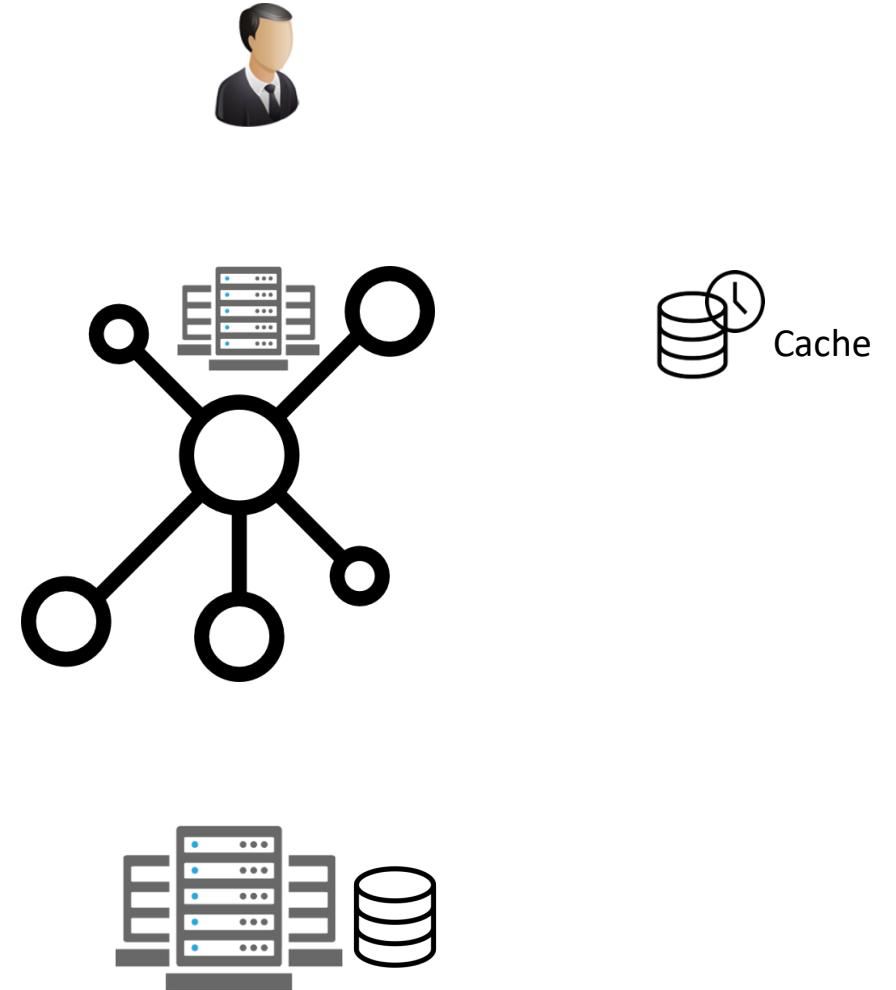
A – Deployment Location

- Definition
 - Host of the cache
- Values
 - Client
 - Server
 - + Resource sharing



A – Deployment Location

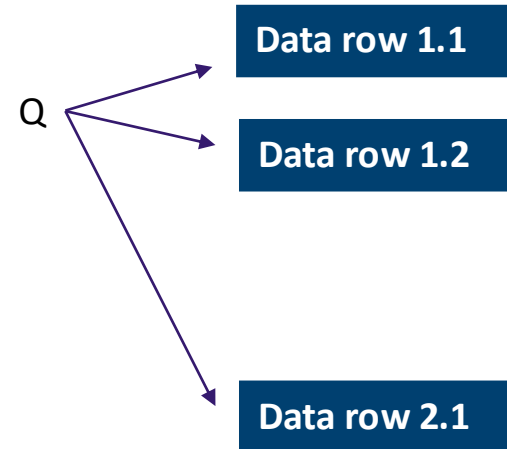
- Definition
 - Host of the cache
- Values
 - Client
 - Server
 - Proxy



A – Deployment

Cache entry

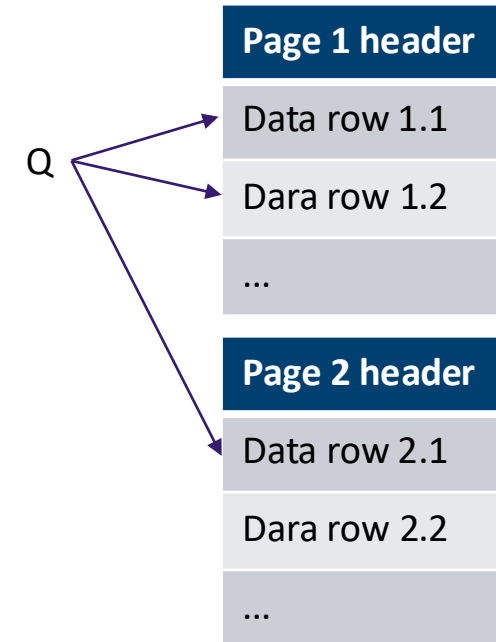
- Definition
 - Granularity of an element
- Values
 - Object [DFM90]
 - + Storage management
 - + Concurrency management



A – Deployment

Cache entry

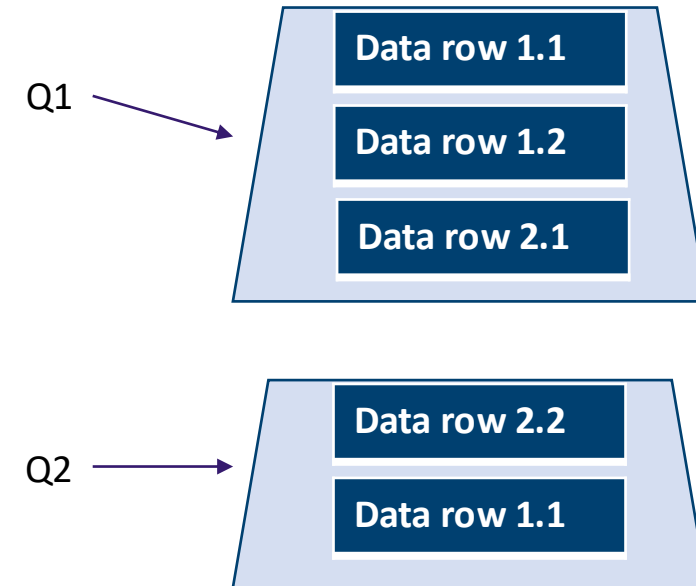
- Definition
 - Granularity of an element
- Values
 - Object [DFM90]
 - Page / fixed size [DFM90]
 - + Aggregation
 - + Prefetching in case of spatial locality



A – Deployment

Cache entry

- Definition
 - Granularity of an element
- Values
 - Object [DFM90]
 - Page / fixed size [DFM90]
 - Semantic region / segment [DFJ96]
 - + Aggregation
 - + Storage management
 - + Prefetching in case of semantic locality
 - Concurrency management



B. Basic functionalities

Research

- Objective
 - Looking for a given element in the cache
- Data structures
 - List
 - Table
 - Hash table
 - Etc.

B. Basic functionalities

Replacement

- Objective
 - Choosing elements to remove from the cache
- Some replacement policies
 - FIFO
 - LIFO
 - LRU
 - Etc.
- Importance of the locality
 - Temporal
 - Spatial
 - Semantic

B. Basic functionalities

Resolution

- Objective
 - Retrieving an element not present in the cache
- Consideration
 - Data source(s)
 - Communication protocol
 - Security

C. Optional functionalities

- Prefetching
- Admission
- Negative caching
- Allocation
- Consistency and concurrency

Outline

- I. Motivation and principles
- II. Concepts
- III. Semantic caching
 - A. State of the art: semantic management
 - B. State of the art: content management
- IV. Cooperative caching

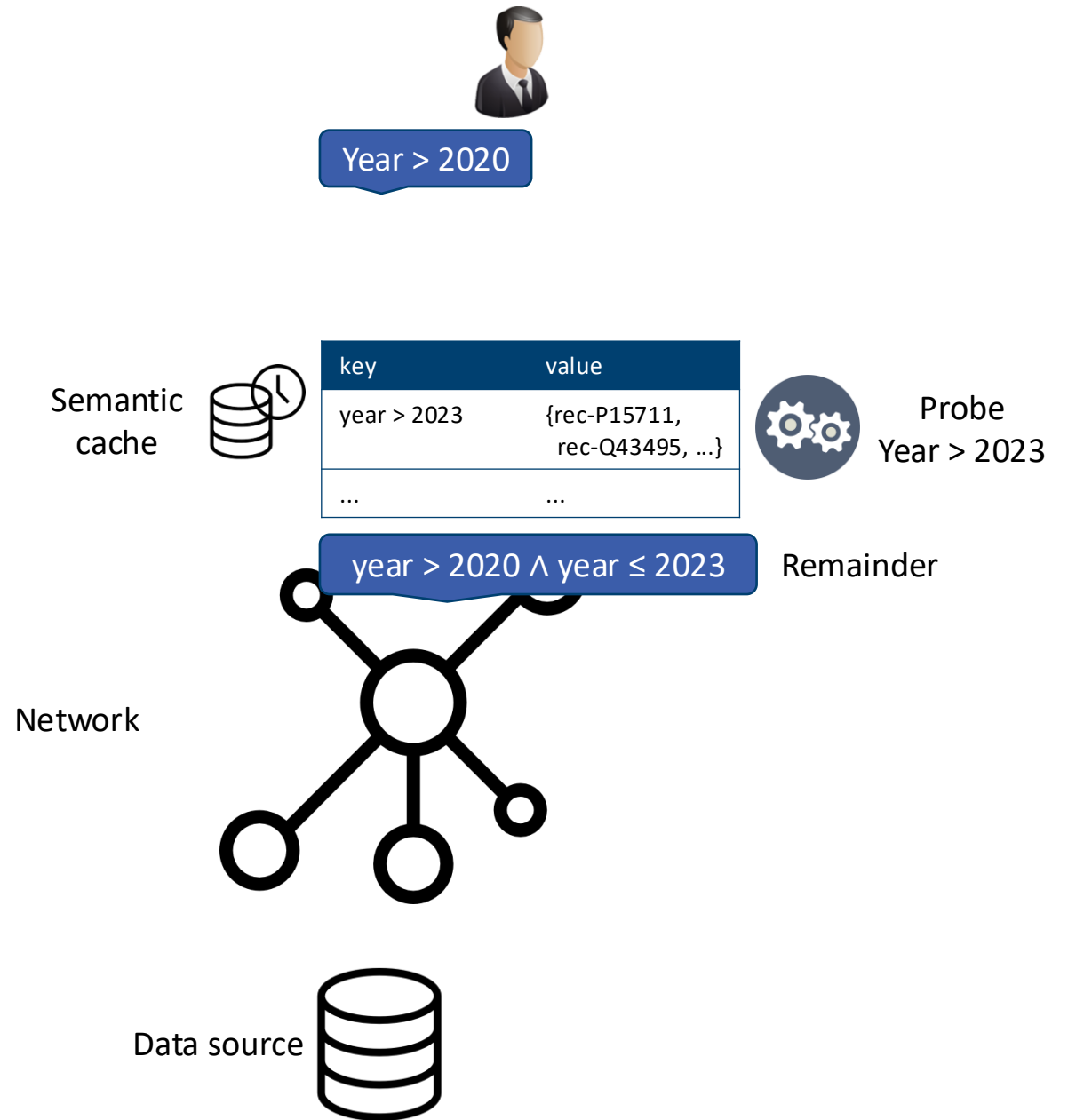
Semantic caching [KB96] [DFJ96]

- Objective

- Taking advantage of local evaluation capabilities
 - Load reduction on servers
 - Data transfers reduction
 - Availability increase

- Principle

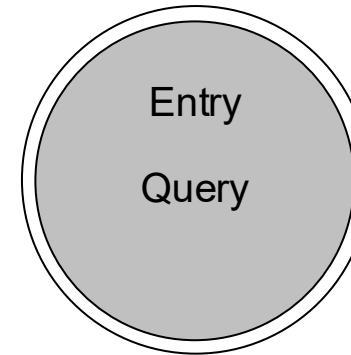
- Management of query results
- Decomposition of posed queries
 - Probe query
 - Remainder query



A. State of the art on semantic caching

Semantic management (1)

- Evaluation
 - Select [DFJ96]
 - Project [RD03]
 - Joint [KB96]
 - Location/spatial query [ZL01]
- Analysis [GG97]
 - Equivalence



Equivalence $\rightarrow$ extended hit

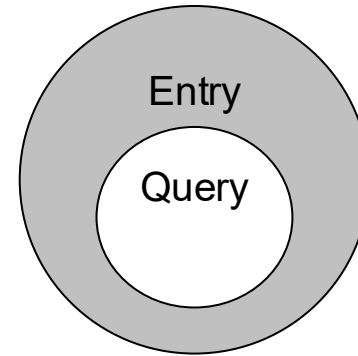
E: year = 2006

Q: year < 2007 $\wedge$ year > 2005

A. State of the art on semantic caching

Semantic management (2)

- Evaluation
 - Select [DFJ96]
 - Project [RD03]
 - Joint [KB96]
 - Location/spatial query [ZL01]
- Analysis [GG97]
 - Equivalence
 - Inclusions
 - Q in E



Query in entry $\rightarrow$ extended hit

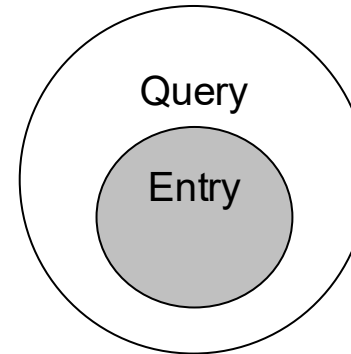
E : year = 2006

Q : year=2006 $\wedge$ auteur= "Blanchet"

A. State of the art on semantic caching

Semantic management (3)

- Evaluation
 - Select [DFJ96]
 - Project [RD03]
 - Joint [KB96]
 - Location/spatial query [ZL01]
- Analysis [GG97]
 - Equivalence
 - Inclusions
 - Q in E
 - E in Q



Entry in query → partial hit
E: year = 2006
Q: year > 2005

A. State of the art on semantic caching

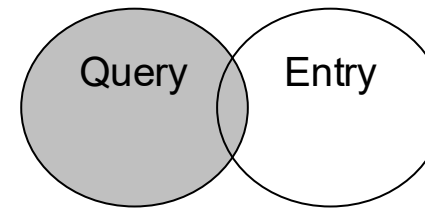
Semantic management (4)

- Evaluation

- Select [DFJ96]
- Project [RD03]
- Joint [KB96]
- Location/spatial query [ZL01]

- Analysis [GG97]

- Equivalence
- Inclusions
- Partial overlap



Partial overlap → partial hit

E: author = "Blanchet" $\wedge$ year = 2006

Q: author = "Blanchet" $\wedge$ species =
virus

B. State of the art on semantic caching

Content management (1)

- Query result [DFJ96]
 - Aggregation of objects
 - Duplication in the cache
 - Combination computation and data

Query result cache

key	value
year=2006	{rec-P15711, rec-Q43495}
year=2010	{}
species=virus	{rec-P15711, rec-P13813}
author=Blanchet	{rec-P13813, rec-P19084}

B. State of the art on semantic caching

Content management (2)

- Predicate and objects [KB96]
 - No duplication in the cache
 - No aggregation of objects
 - Strong consistency predicates/objects
⇒ Combination computation and data

Predicate and objects cache	
key	value
year=2006	{P15711, Q43495}
year=2010	{}
species=virus	{P15711, P13813}
author=Blanchet	{P13813, P19084}

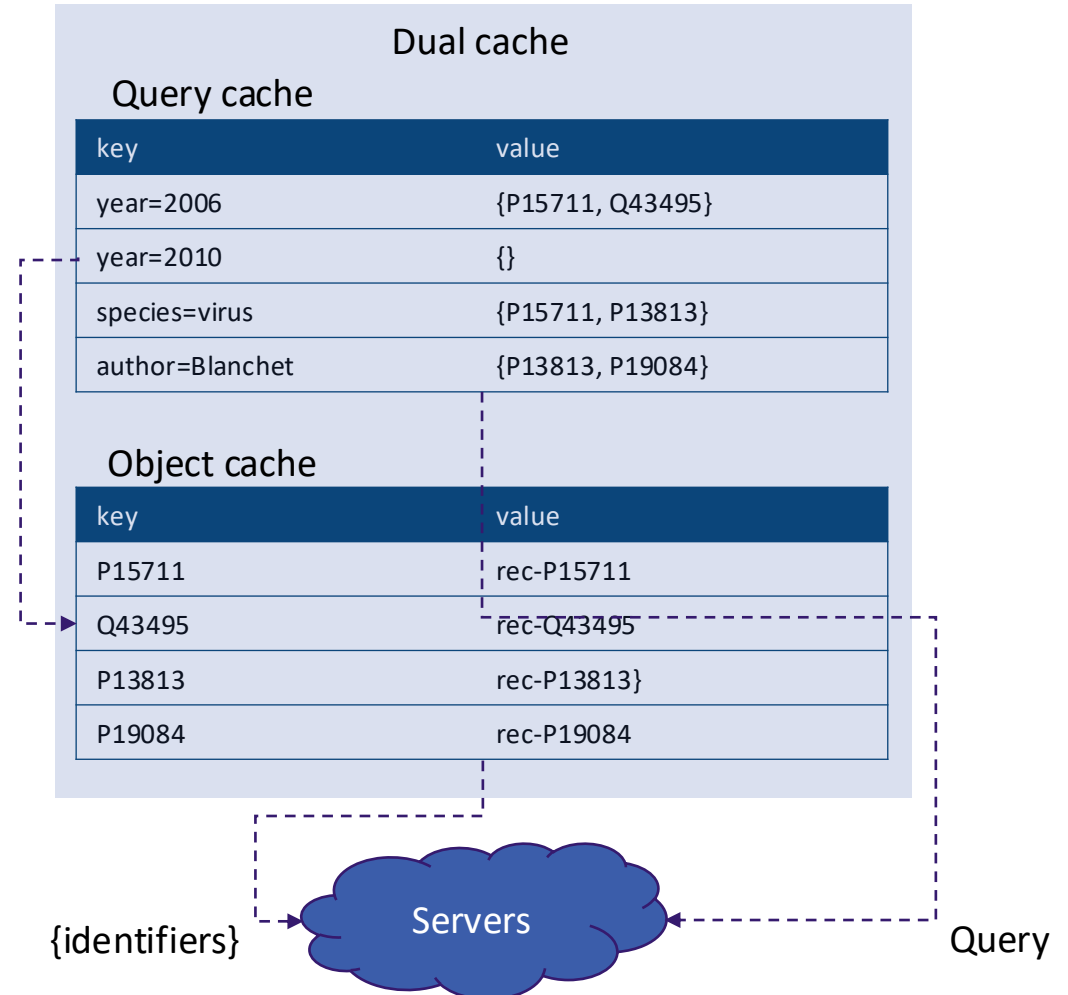
↓

key	value
P15711	rec-P15711
Q43495	rec-Q43495
P13813	rec-P13813}
P19084	rec-P19084

B. State of the art on semantic caching

Content management (3)

- Dual cache [DRL08]
 - Independent caches
 - Query cache $\leftrightarrow$ evaluations
 - Object cache $\leftrightarrow$ data
 - Advantages
 - Evaluations without objects
 - No replication of objects
 - Fine tuning (caches' policies)
 - Limits
 - No aggregation
 - Access by query and identifiers



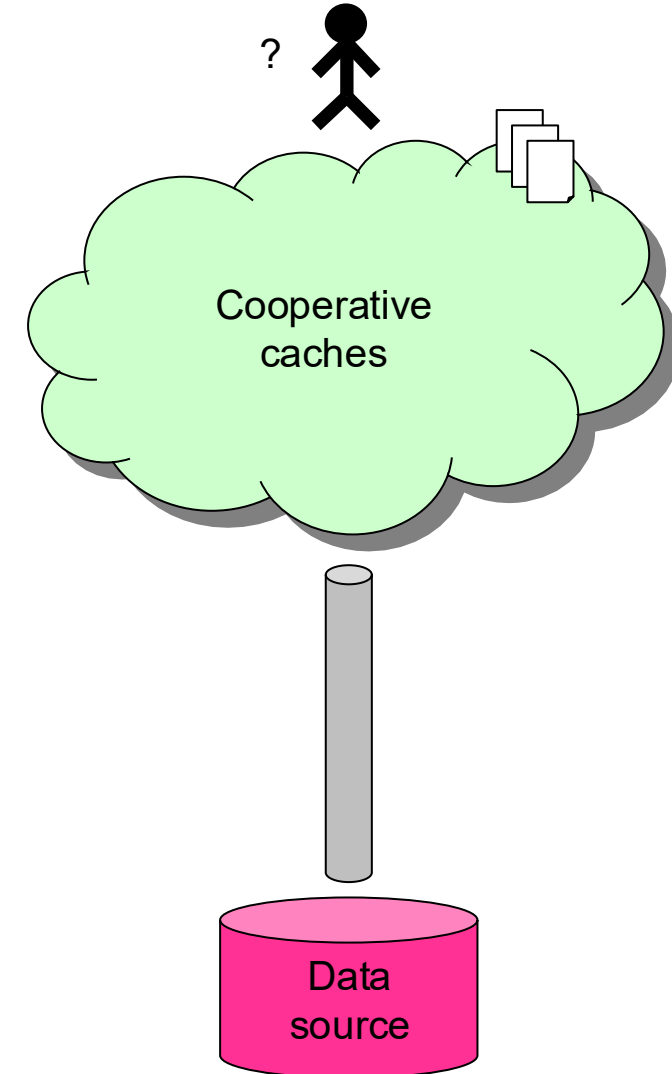
Outline

- I. Motivation and principles
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- III. Semantic caching
- IV. Cooperative caching
 - A. Distributed caching
 - B. Vertical resolution
 - C. Horizontal resolution

Cooperative caching

[BO00] [DMW94] [CDN96]

- Objective
 - Profiting of other caches' resources
- Advantages
 - Data transfers balancing
 - Load reduction on servers
 - Availability's increase



A. Distributed caching [BC95]

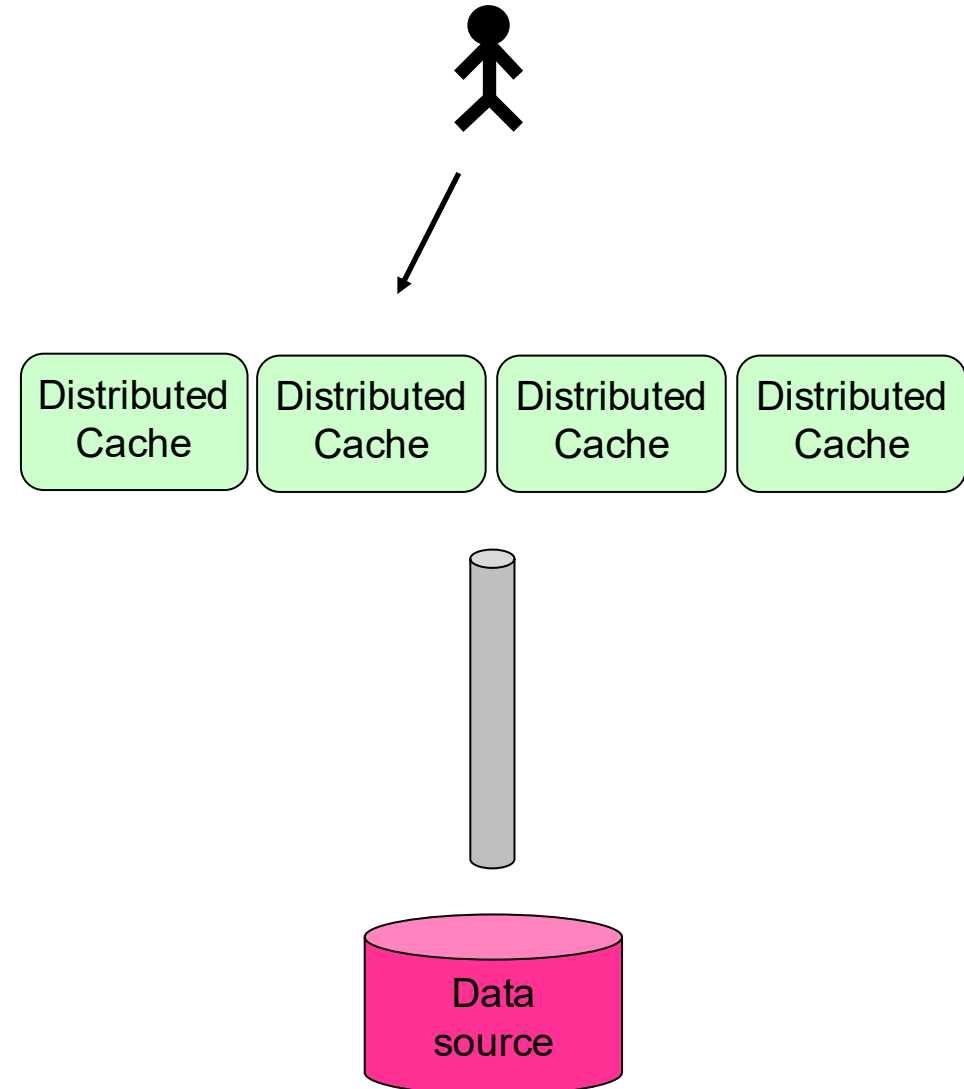
- Principle

Distribution of queries

- Random
- Round robin
- Query-based
- Client-based
- Etc.

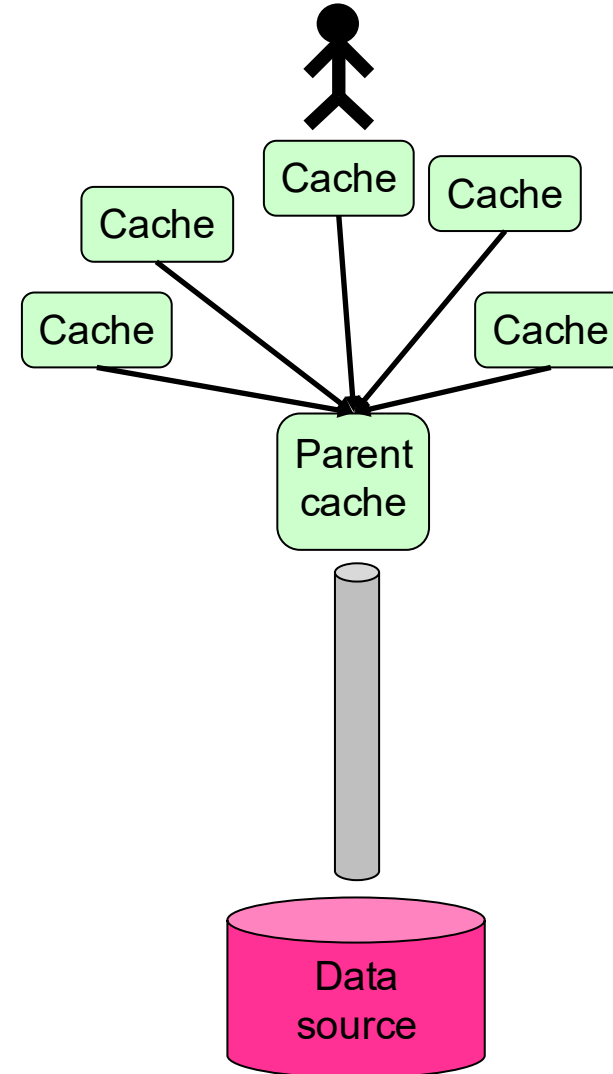
- Drawback

Physically nearby caches



B. Vertical resolution [CDN96]

- Principle
 - Resolution by parents
- Drawback
 - Load on parents
 - Sensitivity



C. Horizontal resolution

- Principle

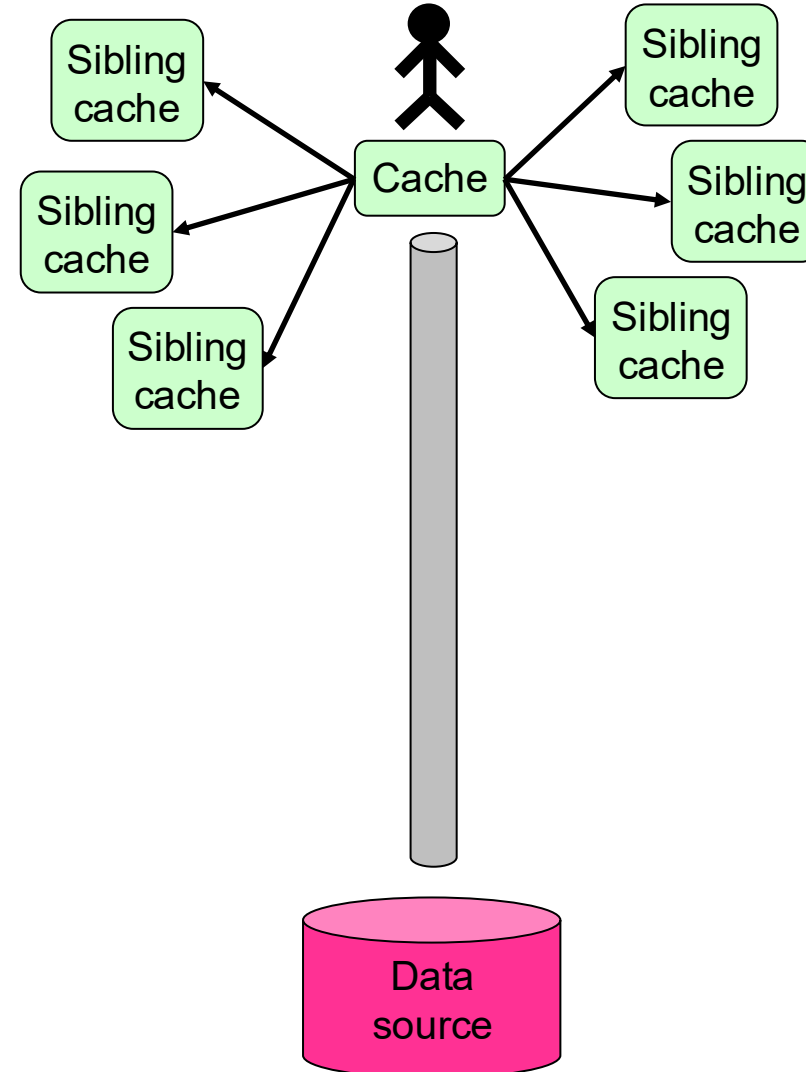
Resolution by siblings

- Protocols

- Flooding [CDN96]
- Catalogue [FCA98]

- Drawback

Large scale management



Questions ?

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